SMART SAFETY COMPANION

Introduction

Smart Safety Companion is a mobile safety application developed using MIT App Inventor. The app continuously monitors the user's phone sensors and automatically sends an emergency alert message to predefined contacts when a strong shake is detected. The alert includes the user's location (latitude & longitude), current direction, and time, and is also displayed inside the app for confirmation.

Objective of the Application

- To provide quick emergency assistance without manual interaction
- To automatically detect danger situations using motion (shake detection)
- To send SOS alerts to trusted contacts
- To display real-time sensor data and emergency status inside the app

Components Used

User Interface Components

Buttons: Start Monitoring, Stop Monitoring

Labels: Display sensor values, status, direction, location, and emergency alert

Vertical Arrangement: Organizes UI components over the background image

Non-Visible Components

AccelerometerSensor – Detects shake/motion

LocationSensor – Gets latitude and longitude

OrientationSensor – Detects phone direction (North, East, South, West)

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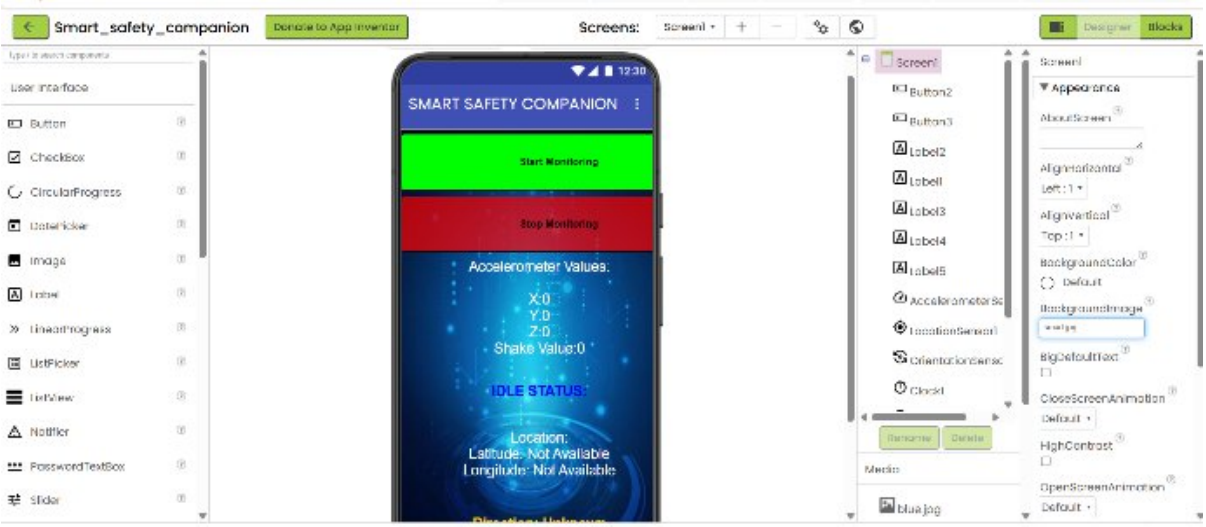
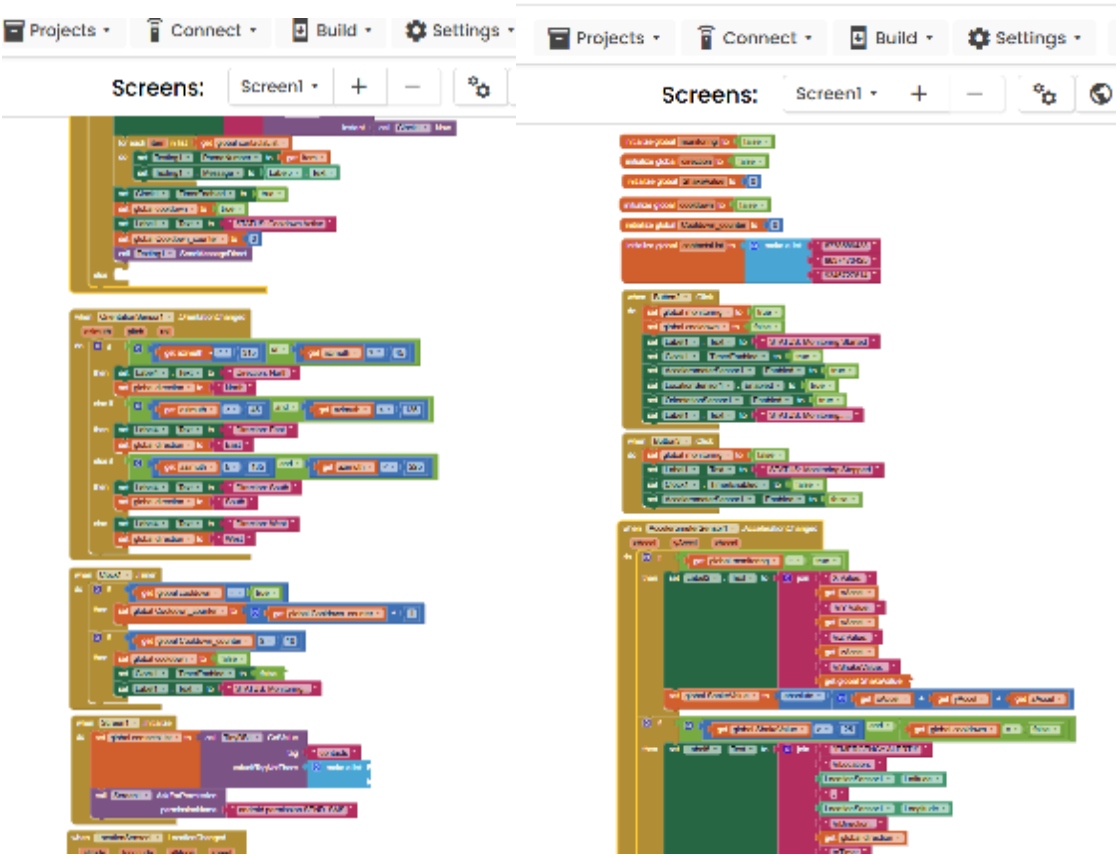
Clock – Used for cooldown timing and time display

Texting – Sends SMS alerts

TinyDB – Stores emergency contact numbers





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App Interface Design

A safety-themed image is set as the Screen background

All buttons and labels are placed inside a Vertical Arrangement so they appear on top of the background image

Color coding is used:

- Green: Start Monitoring
- Red: Stop Monitoring / Emergency Alert
- Blue/Yellow: Status and direction display

Global Variables Used

Monitoring – Indicates whether monitoring is active or not

ShakeValue – Stores calculated shake intensity

Direction – Stores current phone direction

Cooldown – Prevents repeated SMS alerts

Cooldown_counter – Tracks cooldown time

ContactsList – Stores predefined emergency contact numbers

Start Monitoring Process

When the Start Monitoring button is clicked:

Monitoring is enabled

Accelerometer, Location, and Orientation sensors are activated

Clock timer is enabled

Status label is updated to Monitoring Started

This allows the app to continuously track motion and location.

Stop Monitoring Process

When the Stop Monitoring button is clicked:

Monitoring is disabled

All sensors are turned off

Clock timer is stopped

Status label is updated to Monitoring Stopped

This prevents further sensor readings and SMS alerts.

Shake Detection Logic

The AccelerometerSensor continuously reads X, Y, and Z values

ShakeValue is calculated using the absolute sum of these values

If ShakeValue exceeds a defined threshold (e.g., 25), it is considered an emergency shake

This method ensures accidental small movements do not trigger alerts.

Emergency Alert Generation

When a strong shake is detected:

The app checks if cooldown is inactive

An emergency message is created containing:

Emergency alert text

Current latitude and longitude

Detected direction

Current time

The message is displayed in Label5 inside the app

Sending SMS to Emergency Contacts

The app loops through the predefined contact list

The same emergency message is sent to all contacts automatically

No manual contact selection is required

This ensures fast alert delivery during emergencies.

Cooldown Mechanism

After sending an alert, cooldown is activated

Clock timer counts cooldown duration

During cooldown, no new alerts can be triggered

Once cooldown ends, monitoring resumes normally

This prevents SMS spamming due to continuous shaking.

Direction Detection

OrientationSensor reads azimuth value

Direction is determined based on angle ranges:

315°–45° North

45°–135° East

135°–225° South

Else West

The detected direction is shown on the screen and included in the alert.

Location Tracking

LocationSensor continuously updates latitude and longitude

Latest values are displayed in the app

Same values are included in the emergency SMS

Permissions Handling

On app start, SMS permission is requested automatically

Permission is mandatory for sending emergency messages

Output and Result

Real-time sensor values are displayed

Emergency alerts are automatically sent on shake detection

Alert details are shown clearly inside the app

The app works without internet (SMS-based)

Conclusion

The Smart Safety Companion application provides an efficient and reliable emergency alert system using mobile sensors. By combining shake detection, GPS tracking, direction sensing, and automatic SMS alerts, the app ensures quick assistance during dangerous situations. The system is user-friendly, responsive, and suitable for personal safety applications.